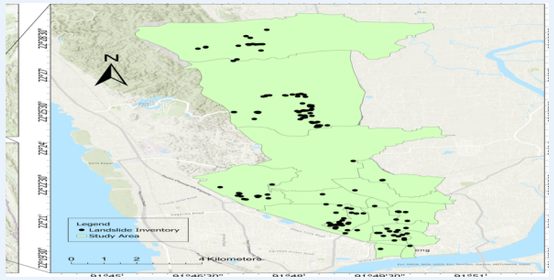
Bayesian benchmark-to-deployment workflow for landslide susceptibility mapping

Chittagong Metropolitan Area, Bangladesh | benchmark all models under one protocol, deploy only the best Bayesian model to preserve uncertainty information

1 Study area and inputs

Study area: Chittagong Metropolitan Area



155 landslide and 214 non-landslide samples.
Factors span terrain, hydrology, climate, land cover, and road-river proximity under a stratified 80/20 split.

2 Benchmark workflow

All four models use the same validation framework

BLR | GPC | LR-L2 | RF

Repeated CV: 5 x 3 | Holdout: 20% | Spatial block CV

Calibration | Friedman p = 0.0005

Best overall benchmark
RandomForest
MCC 0.611 | ROC-AUC 0.866

Best Bayesian candidate
GPC
MCC 0.481 | ECE 0.084

Friedman statistic = 17.727273 | significant performance differences among models

3 Bayesian-only deployment

Weighted ranking across CV, holdout, spatial MCC, Brier, and ECE

Decision rule
Final map export is restricted to Bayesian models only.

Selected for deployment
GPC | Overall rank #2
Chosen to retain model-based uncertainty.

4 Deployed map products

Deployed GPC map: Very High susceptibility covers 9.47% of the area, captures 78.71% of landslides, and yields SCAI = 0.12.

Hazard-screening message

1. Benchmarking identifies RF as the strongest discriminator.
2. Bayesian policy deploys GPC for susceptibility plus uncertainty mapping.
3. High-risk zones are concentrated into a limited spatial footprint.

